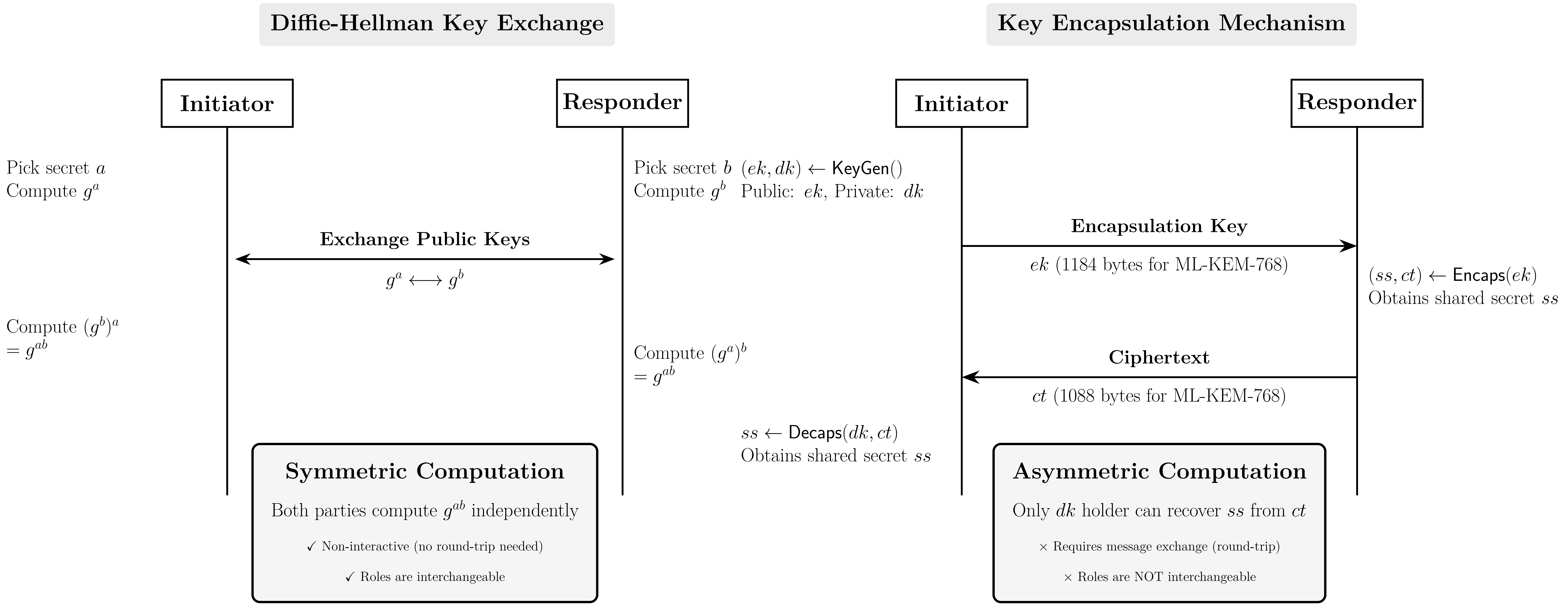
Diffie-Hellman Key Exchange
Key Encapsulation Mechanism
Initiator
Responder
Initiator
Responder
Pick secret a
Compute g^a
Pick secret b
Compute g^b
$(ek, dk) \leftarrow \mathsf{KeyGen}()$
Public: ek, Private: dk
Exchange Public Keys
$g^a \longleftrightarrow g^b$
Encapsulation Key
ek (1184 bytes for ML-KEM-768)
$(ss, ct) \leftarrow \mathsf{Encaps}(ek)$
Obtains shared secret ss
Compute $(g^b)^a$
$= g^{ab}$
Compute $(g^a)^b$
$= g^{ab}$
Ciphertext
ct (1088 bytes for ML-KEM-768)
$ss \leftarrow \mathsf{Decaps}(dk, ct)$
Obtains shared secret ss
Symmetric Computation
Both parties compute g^{ab} independently
✓ Non-interactive (no round-trip needed)
✓ Roles are interchangeable
Asymmetric Computation
Only dk holder can recover ss from ct
× Requires message exchange (round-trip)
× Roles are NOT interchangeable